Data Science Project Proposal

Outcome 1: Team Collaboration & Project Planning

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1. Project Title

"Do low-rated chess players perform better with uncommon openings or common openings?"

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2. Team Information & Roles

Project Team:

Each team member should have a defined role and responsibilities.

Name	**Role(s)**	**Responsibilities**
Anna Bekesi	Analyst	Creating python scripts that interrogates our chess data and provides insights that help discover whether or not we can confidently reject our null hypothesis
James Murphy	Team leader	Overseeing the entirety of the project, contributing to all tasks as needed
Margaret Morgan	Quality Control and documentation coordinator	Checking whether the project is on course and makes sense, and that each stage of the project is being correctly documented.
Callum Rennie	Data Visualization	Creating visualisations and dashboards that allow us to gain insights into our data and present our findings to an audience

Collaboration Tools:

- **Communication:** Microsoft Teams
- **Version Control & Documentation:** Git, GitHub, OneDrive.
- **Project Management:** Microsoft Planner.

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3. Problem Statement & Project Scope

- **Problem Definition:**

  Chess Grandmaster Ben Finegold has strongly contended that studying chess openings (the first few moves of a game) and opening choice is completely irrelevant for low-rated players (under 2000 Elo). However, this claim has not been rigorously tested with empirical data. This project aims to investigate whether playing uncommon openings (used in less than 20% of games) affects performance compared to playing common openings.

  If a significant difference in win rates is found between the two groups, it would indicate that opening choice does impact performance at lower levels, contradicting Finegold's assertion.

  Conversely, if no significant difference is found, this would suggest that opening choice does not play a measurable role in performance at lower levels, aligning with Finegold's assertion. However, failing to reject the null hypothesis does not prove that opening choice is irrelevant—it only indicates that our data does not provide strong evidence of an effect.

- **Statistical Analysis Plan:**

  To determine whether opening choice impacts performance, we will conduct a two-tailed binomial test comparing win/loss rates between players who use uncommon openings (<20% of games) and those who use common openings. This test will allow us to assess whether there is a statistically significant difference in win rates between the two groups.

  Null Hypothesis (H_0): No difference in win rates between common and uncommon opening players.

  Alternative Hypothesis (H_1): A significant difference exists in win rates (either favouring common or uncommon openings).

  Significance Level (α): 0.05

  Exclusion of Draws: Since binomial tests require binary outcomes, draws will be excluded from the dataset.

  Additional Considerations: We will check assumptions such as sample size adequacy and potential confounders (e.g., rating differences).

- **Significance:**

  This study is important for both the chess community and the broader field of data-driven decision-making:

  Challenging Expert Opinions—Grandmaster Ben Finegold’s assertion that opening study is irrelevant for low-rated players is widely discussed but lacks empirical testing. By providing data-driven insights, this research adds clarity to an ongoing debate in the chess community.

  Practical Guidance for Low-Rated Players—Many beginners and club players struggle with how much time to spend on opening study. If results show a performance difference between common and uncommon openings, it will help players make more informed training decisions.

  Implications for Chess Coaching & Training—Coaches often advise students to stick to common openings, but if uncommon openings provide an advantage, this could shift training recommendations.

  A Case Study in Data Science & Statistical Analysis—Beyond chess, this project applies real-world data science methods, making it a valuable case study in predictive modelling and hypothesis testing. The methodology could be applied to other domains where expert opinions conflict with empirical evidence.

- **Stakeholders:**

  Chess learners and enthusiasts who want to improve their gameplay.

  Coaches and trainers looking for data-driven insights on opening strategies.

  The broader chess community, including online platforms such as Lichess and Chess.com.

- **Scope:**

  Included:

  > Analysis of Lichess game data, categorization of openings, statistical comparison of performance metrics.

  Excluded:

  > Psychological factors influencing chess improvement, individual playing styles, long-term improvement tracking.

Key Questions:

- Does the choice between uncommon and common openings impact the performance of low-rated players?
- How significant is the difference in performance between players who play uncommon vs. common openings?
- What factors (e.g., time control, rating differences) influence the results?
- How do we ensure a fair comparison between groups?

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4. Objectives & Key Milestones

Project Objectives

- Collaborate effectively within a data science team.
- Develop a structured project plan aligned with the **PPDAC Data Science Lifecycle**.
- Ensure ethical and legal compliance in data handling.
- Set clear deliverables and success criteria.

Key Milestones & Timeline

Milestone	**Task Description**	**Start Date**	**End Date**	**Responsible Member(s)**
		DD/MM/YYYY	DD/MM/YYYY	[Name]
		DD/MM/YYYY	DD/MM/YYYY	[Name]
		DD/MM/YYYY	DD/MM/YYYY	[Name]
		DD/MM/YYYY	DD/MM/YYYY	[Name]

5. Ethical Considerations & Compliance

- **Data Bias:** We will ensure fairness by analyzing data across various rating ranges and avoiding misleading conclusions.
- **Privacy & Security:** Adhere to GDPR and Lichess data usage policies, ensuring no personally identifiable information is used.
- **Transparency:** Document all decisions regarding data usage, methodology, and model selection for reproducibility.

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6. Deliverables & Submission Timeline

Deliverable	**Description**	**Due Date**
Project Proposal	Define business problem, dataset, and methodology	DD/MM/YYYY
Project Plan	Detailed team roles, milestones, and collaboration strategy	DD/MM/YYYY